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Social media use, mental health and sleep: A systematic review with meta-analyses

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ABSTRACT

Background: The literature investigating the relationship between social media use, mental health, and sleep has produced inconsistent findings. Younger people spend more time on social media than other age groups, and are more likely to be impacted by social media use. This systematic review with meta-analysis aimed to synthesise the evidence on the associations between social media use, mental health, and sleep of young individuals.

Methods: Electronic databases PubMed, Scopus and PsycINFO were searched using an established methodology and pre-determined search terms for studies that reported the association between social media use, mental health, and sleep.

Results: The search yielded 6108 articles, of which 182 (n = 1,169,396) were eligible for the systematic review, and 98 (n = 102,683) could be included in the meta-analyses. The systematic review identified a high level of heterogeneity in the study results. Meta-analyses found small but significant positive associations between social media use, depression, and anxiety. In addition, problematic social media use was positively associated with depression, anxiety, and sleep problems, and negatively associated with wellbeing. Geographical location, anxiety measure type, study design, age, and gender were identified as potential moderators.

Limitations: Associations for specific social media platforms as well as some moderator effects were not examined due to an insufficient number of studies.

Conclusions: This study provides important evidence of an association between social media use/ problematic social media use, mental health, and sleep. The findings support future longitudinal research to identify the directions and underlying mechanisms of the inter-relationship between these variables.

1. Introduction

In this era of information and technology, social media platforms are a popular means of online social engagement. Social media use (SMU) encompasses online activities on platforms designed for social interaction, involving content sharing, networking, communication, and engagement with others, serving various personal, social, and professional purposes. Across the globe, almost five billion people (59.4 % of the world population) are active social media users and spend more than two and a half hours on social media daily (Kemp, 2023). Young adults in their twenties make up one-third of total social media users, and users between 16 and 24 years old spend more time on social media daily than any other age group (3 h 11 min for females, 2 h 46 min for males).

From the beginning of the last decade, behavioural scientists have been concerned about the possibility of maladaptive use of social media, often referred to as 'social media addiction' (SMA) or 'problematic social media use' (PSMU). In the literature, the terms 'SMA' and 'PSMU' are often used interchangeably to refer to the phenomenon of excessive SMU characterized by either addiction-like symptoms (e.g., salience, withdrawal symptoms, mood regulation, etc.) and/or reduced self-regulation associated with poorer mental health (Ahmed et al., 2022; Bányai et al., 2017; Casale et al., 2018; Marino et al., 2018). In published studies on the topic, SMA is the most widely used term followed by PSMU (Pellegrino et al., 2022). However, SMA has not been formally recognized as a disorder in the Diagnostic and Statistical Manual of Mental Disorders – 5 [DMS-5] (American Psychiatric Association, 2013).

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Moreover, its use may detract from the arguably more serious substance addictions (Carbonell and Panova, 2017). Therefore, the term 'PSMU' is often favoured over 'SMA'. PSMU is a broad term that incorporates excessive SMU with addiction-like symptoms, which limits users' daily activities and is negatively associated with mental health (Andreassen and Pallesen, 2014; Griffiths, 2005; van den Eijnden et al., 2021). In a recent study, Finserås et al. (2023) explored PSMU as a multidimensional concept that includes the subjective overuse of social media, accompanied by social obligations (feeling the need to present on social media) and a source of concern, with these also being associated with mental health problems. In the present study, we therefore opted for the use of the term 'PSMU'.

Both adolescents and young adults are more sensitive than older age groups to the benefits and harms of SMU (Crone and Konijn, 2018; Hayes et al., 2015). SMU can facilitate the expression of one's opinions, thoughts, and feelings, as well as increase social connectedness and reduce loneliness (Deters and Mehl, 2012; O'Keeffe et al., 2011). On the other hand, SMU and PSMU are associated with depression, anxiety, psychological distress, lower life satisfaction, poor mental health, and eating disorders (Bányai et al., 2017; Best et al., 2014; Keles et al., 2020; Marino et al., 2018; Rodgers et al., 2020; Shensa et al., 2017). Several systematic reviews, meta-analyses, umbrella reviews, and narrative review have been conducted on the association between SMU/PSMU and mental health [see (Appel et al., 2020; Arias-de la Torre et al., 2020; Cunningham et al., 2021; Fumagalli et al., 2024; McCrae et al., 2017; Schønning et al., 2020; Seabrook et al., 2016; Valkenburg et al., 2022)]. Published reviews have reported weak and/or inconsistent associations between SMU and mental health as well as higher heterogeneity in the strength of their associations [see (Best et al., 2014; Ivie et al., 2020; Keles et al., 2020; Liu et al., 2022; Sanders et al., 2024; Santos et al., 2023; Ulvi et al., 2022; Yin et al., 2018)]. In contrast, the association between PSMU and mental health has been found to be consistent and significant (Huang, 2022; Lopes et al., 2022; Marino et al., 2018; Piteo and Ward, 2020; Shannon et al., 2022). However, available meta-analyses have several limitations including a focus restricted to the association between SMU/PSMU and depression, the combination of several types of mental health outcomes into a single outcome, a relatively small number of studies available for inclusion, and a lack of consideration of study quality and publication bias. Furthermore, findings remain inconclusive as some meta-analyses reported negative associations between SMU and depression (Ma, 2022; Wong et al., 2022), while others reported positive associations (Cunningham et al., 2021; Ivie et al., 2020; Vahedi and Zannella, 2021; Yang et al., 2022; Yoon et al., 2019).

The link between SMU/PSMU and mental health is complex and multifaceted. Several factors may contribute to this relationship, with sleep being potentially one of them. Restful sleep is important for the physical and psychological development of young people. Shorter sleep duration, sleep disturbance, and poor sleep quality are common among adolescents worldwide (Gradisar et al., 2011). The National Sleep Foundation (2011) found that 61 % of 13–18 year olds and 23 % of 19–29 year olds slept <8 h. Sleep problems are associated with mental health and behavioural difficulties among adolescents (Morales-Muñoz and Gregory, 2023). Studies also suggest an association between SMU/PSMU and sleep outcomes, including sleep quality, sleep duration, insomnia, and daytime sleepiness (Das-Friebel et al., 2020; Kelly et al., 2018; Li et al., 2017; Nasirudeen et al., 2017; Wong et al., 2020). However, it remains unclear whether SMU/PSMU causes poor sleep quality, if poor sleep quality causes SMU/PSMU, or if there is a bidirectional association between SMU/PSMU and sleep quality. To date, two systematic reviews have assessed the relationship between SMU and sleep (Alonzo et al., 2021; Kaur et al., 2021). In their review, Alonzo et al. (2021) reviewed 42 studies and reported a significant association between SMU and sleep problems. Kaur et al. (2021) reviewed 45 studies and highlighted some limitations of the past research, particularly their narrow focus on SMU behaviours and sleep dimensions. However, none of these reviews conducted meta-analyses on the associations between all these variables.

The aim of the present study was therefore to systematically review the literature, and conduct a meta-analysis of the associations between SMU/PSMU, mental health, and sleep in adolescents and young adults to address the limitations and gaps in knowledge reviewed above. Specific objectives were i) to assess the associations between SMU/PSMU and mental health; ii) to assess the associations between SMU/PSMU and sleep; iii) to investigate whether sleep mediates or moderates the relationship between SMU/PSMU and mental health; iv) to assess the differences in associations across study regions, study designs, type of social media platforms, measures of social media use, mental health, and sleep; and v) to assess the moderating effects of gender, age, sample size, and study quality on these associations.

2. Methods

This study was prospectively registered on PROSPERO database (CRD42021233495) and followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page et al., 2021).

2.1. Search strategy

PubMed, Scopus and PsycINFO were searched on with the following search string: ((“Social media” OR “Social Network*” OR “Facebook” OR “Instagram” OR “WhatsApp” OR “Twitter” OR “Tiktok” OR “WeChat” OR “Myspace” OR “QQ”) AND (“Us*” OR “Addict*” OR “Problem*”)) AND (((“Sleep”) AND (“quality” OR “pattern” OR “latency” OR “duration” OR “problem” OR “Insomnia” OR “Daytime sleepiness” OR “Disturb*” OR “disruption” OR “timing”)) OR (“Mental health” OR “wellbeing” OR “wellbeing” OR “Psychological distress” OR “Positive affect*” OR “Negative affect*” OR “depressi*” OR “anxiety” OR “Eating disorder*” OR “bulimi*” OR “anorexi*”)) AND (“juvenile” OR “minor” OR “child*” OR “tween*” OR “teen*” OR “adolescen*” OR “young adult*” OR “student*” OR “youth” OR “young”).

2.2. Screening

Duplicate records were first removed. Next, following a previously established methodology (Cherbuin et al., 2015), a single author (OA) screened entries by title. The abstracts of the remaining entries were then double-screened (OA, EW, AD, and NC) against inclusion and exclusion criteria. Disagreements were resolved by consensus. Next, full-text and supplementary materials of the selected studies were obtained and double-screened by the same authors. Bibliographies of retrieved studies and previous reviews (extracted through literature search) were also examined to identify any additional studies which may not have been detected through the database search. EndNote v.20 was used for article screening.

2.3. Inclusion and exclusion criteria

Studies were included if they i) assessed the association between SMU/PSMU and sleep quality/problem, SMU/PSMU and mental health outcomes (e.g., anxiety, depression, wellbeing, and eating disorder) [SMU was assessed through usage duration, frequency, or questionnaires/scales; PSMU, mental health, and sleep assessed using questionnaires/scales]; ii) included children, adolescents, or young adults provided the mean sample age $\leq$ 25 years; iii) included at least 20 participants; iv) used a cross-sectional/longitudinal observational or RCT design; and v) were published in peer-reviewed academic journals; and vii) were written in or translated into English.

Studies were excluded if they: i) focused exclusively on general internet use, problematic internet use, internet addiction, or internet gambling; ii) focused on problematic smartphone use; iii) reported

clinical case reports; iv) exclusively investigated clinical populations (i.e., brain disorders/cognitive disorders, etc.); v) included <20 participants; vi) were published in non-English languages; vii) reported on samples with a mean age above 25 years; viii) were review articles, theses, conference abstracts, or non peer-reviewed literature; ix) had a retrospective design (i.e., studies based on recollection of past feeling, for example, how often one felt down, depressed, or hopeless for a two-week period in six months ago).

2.4. Data extraction

The data were double-extracted by three authors (OA, DEO, and KA). Discrepancies between the authors were resolved by consensus. Data included the following information: i) Basic publication information (title, authors' name, publication year, and country); ii) Study design and the demographic description of participants (sample size, sample source, sampling technique, type of study, age, and gender); iii) Description of measures used for assessing SMU, sleep, and mental health; iv) Major findings; and v) Statistical estimates of the association between SMU, sleep, and mental health (statistical method, correlation coefficient and regression coefficient estimates (β coefficients, odd ratio coefficients, 95 % confidence intervals, and standard errors). Where essential data were missing, authors of the identified manuscripts were contacted to obtain additional information. Where missing information required for meta-analysis could not be sourced, the study contributed to the qualitative review but not to the meta-analysis.

2.5. Outcome measure

Pearson correlation coefficients (r) were considered as measures of association between SMU/PSMU, mental health, and sleep. In studies where they were not available, estimates were computed from standardized and unstandardized β-coefficients, and odd-ratios (OR) using conversion formulas described in Table S1. Standardized β-coefficients were utilized for assessing the longitudinal associations between variables.

2.6. Statistical analysis

Analyses were conducted in R [R Core Team, 2021, version 4.0.5] using RStudio [Posit team, 2023, version 2023.06.1 + 524]. Meta-analyses and meta-regressions were computed with the *metafor* package 3.0–2 (Viechtbauer, 2010) and subgroup analyses used the *meta* package 4.19–0 (Balduzzi et al., 2019).

2.7. Meta-analyses

Where a minimum of three studies was available, meta-analyses were conducted separately for SMU/PMSU, mental health and sleep. Pearson correlation coefficients were pooled using random effects models, as studies were heterogeneous in design and method. We examined heterogeneity using the Cochran *Q* statistic and the I^2 statistic (values of 25 %, 50 %, and 75 % are indicative of low, medium, and high heterogeneity, respectively) (Higgins et al., 2019).

2.8. Meta-regressions

We investigated the moderating effects of age, proportion of females, sample size, publication year, and study quality. Sub-group analyses were run to explore differences in effect size by region (Asia, Europe, North America, and Oceania), study design (cross-sectional and longitudinal), social media platform (General, Facebook, Instagram, and other social media platforms), type of SMU (overall, active, and passive), and measures for assessing study variables. Groups having at least three coefficients were included in subgroup analyses. We did not run meta-regression if the effect size from a meta-analysis was non-significant.

2.9. Quality assessment

The quality of each study was assessed with a modified version of the Newcastle-Ottawa Scale (Wells et al., 2013); Table S2) by two authors. This scale measures the quality of four broad methodological domains – Selection (sample), Exposure (SMU/PSMU assessment), Comparability (confounders), and Outcome (sleep and mental health). Total scores ranged between 0 and 10 with a higher score indicating better quality. Discrepancies in quality scores were resolved by consensus.

2.10. Bias assessment

We assessed publication bias graphically using funnel plots and the Egger regression tests (Egger et al., 1997; Viechtbauer, 2010) with standard error as a predictor to provide objective interpretation. We used the trim-and-fill method (Duval and Tweedie, 2000) to estimate the effect of extreme studies missing from the meta-analyses. The effect of overly influential studies was investigated by conducting sensitivity analyses. Influential studies were identified using the covariance ratio values and plot. A study having a covariance ratio below 1 suggests that removing that study results in a more precise estimate of the effect size (Harrer et al., 2022) (p.160). After excluding influential studies, random effect models were run again to examine whether there was any change in strength and direction of associations between study variables.

3. Results

Fig. 1 shows the PRISMA search and screening process. The search identified a total of 6105 records. After removing duplicates ($n = 1398$), 4707 articles were considered for possible inclusion. After title, abstract and full article screening (179 articles), and bibliography examinination (three articles), a total of 182 articles reporting on 1,169,396 individuals met the inclusion criteria and were included in the systematic review. Some discrepancies between raters occurred at the abstract ($n = 32$) and full article ($n = 29$) screening stages, which were resolved by consensus. Characteristics of the included studies are presented in Table S3. Of these, 98 articles ($n = 102,683$, median = 470, interquartile range (IQR) = 532, weighted mean age = 17.4 years, female = 55.3 %) reported sufficient information and shared compatible methodologies to be included in the meta-analysis. Among the included articles, 139 were cross-sectional [1–139], 41 longitudinal [140–180], one included both cross-sectional and longitudinal designs [181], and one was a cohort study [182]. Six articles recruited only females [58,111,117,153,166,167]. Sample sizes ranged between 63 [142] and 369,595 [171] participants (median = 582).

3.1. Quality assessment

Most studies were rated as being of moderate (39.0 %) to high (46.7 %) quality, with only 14.3 % rated as low quality (Fig. S1). Mean study quality scores (out of 10) ranged from 4.82 (SMU and sleep duration) to 8.03 (PSMU and sleep problems) (Table S4 and Fig. S2). Most studies assessed SMU (70.9 %), followed by depression (53.9 %), PSMU (37.9 %), sleep (26.4 %), anxiety (20.9 %), and wellbeing (15.4 %) (Fig. S3).

3.2. Social media use

Among the studies that assessed SMU, 73.0 % of studies reported on general SMU. Of the studies that reported specific platform use, 17.5 % reported on Facebook use, 10.3 % reported on Instagram use, 7.9 % on Twitter use, and 9.5 % reported on use of other social media platforms (WeChat, Snapchat, WhatsApp, YouTube, Google+, Vine, Tumblr, Pinterest, LinkedIn, and BlackBerry Messenger). Most studies assessed SMU duration (50 %), followed by frequency (24.6 %). SMU was most frequently assessed with ad-hoc questions (68.3 %), and where validated scales were used (31.7 %), the Facebook Intensity Scale (Ellison et al.,

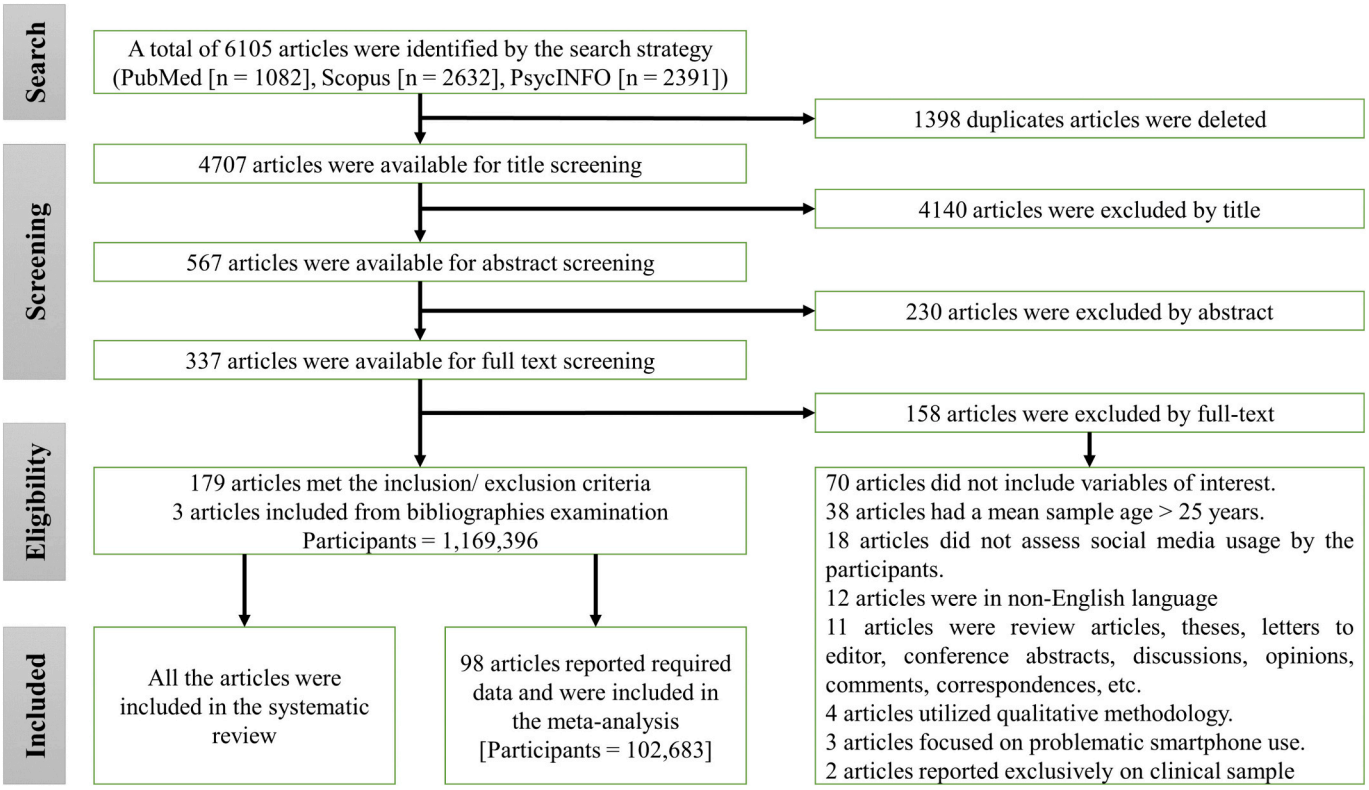


Fig. 1. PRISMA flow diagram presenting the number of studies excluded and included at every stage of the review process, and the final number of studies included in the systematic review and meta-analysis.

2007) was the most common measure (8.7 %). Per day mean SMU duration ranged between 45 and 252 min [74,147]. Percentages of social media users who use social media for five hours or more ranged from 20.8 % to 64.9 % [85,98].

3.3. Mental health

Among studies that assessed depression, 20.4 % of studies assessed depression using the Centre for Epidemiologic Studies of Depression Scale (Radloff, 1977), 11.2 % of studies used the Patient Health Questionnaire 9 (Kroenke et al., 2001), 8.2 % of studies used the Depression Anxiety Stress Scale 21 (Lovibond and Lovibond, 1995) and the rest of the studies used other measures for assessing depression [for details see Table S5]. Among studies that assessed anxiety, 17.9 % of studies used the Depression Anxiety Stress Scale 21 (Lovibond and Lovibond, 1995), 10.3 % of studies used the State-Trait Anxiety Inventory Short form (Marteau and Bekker, 1992), and the rest of the studies used other measures for assessing anxiety. Regarding wellbeing, 14.3 % of studies used composite scores combining the Satisfaction with Life Scale (Diener et al., 1985) and the Positive Affect and Negative Affect Scale (Watson et al., 1988), 10.7 % of studies used the Psychological Wellbeing Scale (Diener et al., 2009), and the rest of the studies used other measures for assessing wellbeing. Concerning eating disorders, 26.6 % of studies that assessed eating disorders used the Eating Disorder Examination Questionnaire (Fairburn and Beglin, 1994) and the rest of the studies used other measures.

3.4. Sleep

Among studies that assessed sleep duration. 55 % of studies assessed sleep duration using two questions (sleep onset time and sleep offset time), 35 % used single questions, 5 % used actigraphy, and the rest of studies did not provide any information about the measure. Among studies that assessed sleep problems, 32.4 % of studies assessed sleep problems using the Pittsburgh Sleep Quality Index (Buysse et al., 1989), 23.5 % of studies used self-developed measures, and the rest of the studies used other measures.

3.5. Social media use and depression

A total of 61 studies assessed the association between SMU and depression. Forty were cross-sectional (65.6 %), 19 were longitudinal (31.1 %), one was a cohort study (1.6 %), and another used both cross-sectional and longitudinal designs (1.6 %). Of those, 28 (45.9 %; representing a combined total sample size of n = 47,494 participants) reported significant cross-sectional associations between SMU and depression [10,11,15,28,31,35,40,41,48,59,66,69,81-84,86,87,92,100, 108,128,135-137,145,159, 182], and 16 studies (26.2 %; representing n = 10,051 participants) reported non-significant associations [4,8,22,27,30,43,44,61,62,93,102,104,117,127, 153,181]. Six studies (9.8 %; n = 17,279 participants) reported significant longitudinal associations between SMU and depression [146,154,158,163,107,180], 8 studies (13.1 %; n = 82,433 participants) did not detect significant longitudinal associations [140,143,150,157,159,164,178,181], and one study (1.6 %; n = 5217 participants) reported mixed results across different platforms [152]. Moreover, two longitudinal studies (3.3 %; n = 3562 participants) found that depression predicted later SMU [154,164], while three studies (4.9 %; n = 4345 participants) did not detect such effects [143,150,157].

Forty studies (n = 47,975, median = 342, interquartile range (IQR) = 506.75, weighted mean age = 17.1 years, female = 55.7 %) could be included in the meta-analysis. In aggregate, there was a small and significant association between SMU and depression ($\bar{r}$ = 0.102, 95 % CI [0.074, 0.130]) (Table 1 and Fig. S4). In sub-analyses excluding very influential studies ($\bar{r}$ = 0.101, 95 % CI [0.077, 0.126]), this association did not differ in strength or direction (Table S6). In addition, baseline

Table 1
Summary of meta-analysis results for association between social media use, mental health, and sleep outcomes.

Variables	k	N	Weighted mean age	ES (95 % CI)	Q	I^2	Egger's test Z	Trim & fill est. (Q)	Trimmed studies
Social media use									
Depression	42 (46)	47,975	17.12	0.102*** (0.074, 0.130)	521.559	86.52	−0.486	0.119 (738.470)	5
Anxiety	18 (19)	24,657	17.82	0.136*** (0.083, 0.188)	241.223	92.0	−0.020	0.188 (489.326)	6
Wellbeing	7	4273	18.45	0.004 (−0.175, 0.184)	115.871	97.01	1.233	0.004 (115.871)	0
Eating disorder	4 (5)	1842	18.05	0.125 (−0.008, 0.258)	42.658	87.38	1.399	0.056 (66.216)	2
Sleep duration	6	14,417	14.66	−0.062 (−0.163, 0.039)	99.068	92.68	−0.002	−0.062 (99.068)	0
Sleep problems	11	19,889	15.54	0.079 (−0.037, 0.194)	254.691	97.93	−0.477	0.079 (254.691)	0
Problematic social media use									
Depression	33	23,966	18.60	0.297*** (0.253, 0.341)	413.993	90.59	1.221	0.240 (709.768)	10
Anxiety	12	10,020	20.64	0.278*** (0.221, 0.336)	64.702	84.50	1.435	0.250 (91.453)	2
Wellbeing	2 (3)	2842	19.59	−0.246*** (−0.352, −0.140)	16.269	87.92	−0.397	0.246 (16.269)	0
Sleep duration	3	1685	20.85	0.038 (−0.010, 0.086)	0.461	0.00	0.675	0.025 (1.568)	2
Sleep problems	10	7967	16.46	0.189** (0.054, 0.323)	258.007	97.06	0.848	0.189 (258.007)	0

** $p < .01$.
*** $p < .001$.

SMU had a significant longitudinal association with increased depressive symptoms ($\bar{r}$ = 0.114, 95 % CI [0.050, 0.178], $p < .001$, Q_6 = 28.356, I^2 = 74.2 %). However, baseline depressive symptoms were not significantly associated with changes in later SMU ($\bar{r}$ = 0.026, 95 % CI [−0.009, 0.061], p = .150, Q_6 = 12.961, I^2 = 88.15 %) (Fig. S5).

Subgroup analyses (Table 2, Figs. S6–S11) showed that the associations between SMU and depression differed significantly by geographical region. On average studies from Oceania reported stronger associations than studies based in Asia, Europe, and North America. This association also significantly differed by social media platforms (general vs specific platforms [Facebook, Instagram, and other specific platforms]) with studies on general SMU and Instagram reporting stronger associations than studies reporting on all SMU platforms, Facebook or other specific social media platforms. However, the strength of the association was not significantly different across study designs (cross-sectional vs longitudinal), depression measures (CESD vs DASS-21 vs PHQ-9 vs others), and SMU measures (validated vs non-validated). Subgroup analyses also showed non-significant differences in the strength of the association between studies including samples aged between 12 and 30 and those including all ages (Table S7). However, there was a moderating effect of age (Table 3) such that weaker associations between social media use and depression were detected in younger participants. However, no moderating effect of gender, sample size, publication year or study quality were detected.

3.6. Social media use and anxiety

Twenty-six studies assessed the association between SMU and anxiety. Among studies, 17 were cross-sectional (60.7 % of 26 studies), eight were longitudinal (28.6 %), and one was a cohort study (3.6 %). Among all the studies, thirteen studies (50 %; representing n = 15,658 participants) reported significant cross-sectional associations between SMU and anxiety [10,23,59,81,103,108,114,128,135,137,145,153,182], while seven studies (26.9 %; n = 3565 participants) reported non-significant associations [4,15,41,93,117,127,136]. Two studies (7.7 %; n = 3367 participants) reported significant longitudinal associations between SMU and later anxiety [163,170] while three studies (11.5 %; n = 11,244 participants) did not detect significant longitudinal associations [150,174,177] and one study (3.8 %; n = 5217 participants) reported both mixed results between use different platforms and later anxiety [152].

Eighteen studies (participant n = 24,657, median = 512, IQR = 383.5, weighted mean age = 17.8 years, female = 56.8 %) could be included in the meta-analysis. In aggregate, there was a small but significant association between SMU and anxiety ($\bar{r}$ = 0.136, 95 % CI [0.083, 0.188]) (Table 1 and Fig. S12). In sub-analyses excluding influential studies ($\bar{r}$ = 0.125, 95 % CI [0.085, 0.164]), this association did not differ in strength or direction (Table S6). In addition, baseline SMU had a significant longitudinal association with increased anxiety symptoms ($\bar{r}$ = 0.147, 95 % CI [0.063, 0.231], $p < .001$, Q_6 = 4.995, I^2 = 25.3 %).

Subgroup analysis results (Table 2, Figs. S13–S17) showed that the associations between SMU and anxiety differed significantly by geographical region. On average studies from North America reported stronger positive associations than studies based in Asia and Europe. This association significantly differed by anxiety measures with studies assessing anxiety with the Depression Anxiety Stress Scale (DASS-21) reporting a weaker association than other scales. However, the strength of association was not significantly different across study designs (cross-sectional vs longitudinal), and SMU measure (validated vs non-validated). Subgroup analyses also showed non-significant differences in the strength of the association between studies including samples aged between 12 and 30 and those including all ages (Table S7). In addition, no moderating effect of age, gender, sample size, publication year or study quality were detected (Table 3).

3.7. Social media use and wellbeing

A total of 25 studies assessed the association between SMU and wellbeing. Twenty were cross-sectional (80 % of 25 studies), four were longitudinal (16 %), and one was cohort study (4 %). Among studies, 16 studies (64 %; representing n = 371,236 participants) reported significant cross-sectional associations between SMU and wellbeing [9,13,14,18,20,21,24-26,39,51,67,71,90,112,133], while five studies (20 %; n = 4687 participants) reported non-significant associations [42,53,63,139,182]. Among longitudinal studies, three studies (12 %; n = 10,208 participants) reported significant longitudinal associations between baseline SMU and later wellbeing [144,175,179].

Seven studies (participant n = 4273, median = 580, IQR = 278, weighted mean age = 18.5 years, female = 59.0 %) could be included in the meta-analysis. In aggregate, there was a small and non-significant association between SMU and wellbeing ($\bar{r}$ = 0.004, 95 % CI [−0.175, 0.184]) (Table 1 and Fig. S18). In sub-analyses excluding influential studies ($\bar{r}$ = −0.068, 95 % CI [−0.190, 0.054]), this association did not differ in strength (Table S6).

3.8. Social media use and eating disorders

A total of seven studies assessed the association between SMU and eating disorders. All the studies (representing n = 3461 participants) reported significant cross-sectional associations between SMU and eating disorders [31,29,101,117,125,136,153]. Four studies (participant n = 1842, median = 332, IQR = 105, weighted mean age = 18.1 years, female = 76.3 %) could be included in the meta-analysis. In

Table 2
Summary of subgroup analyses testing the associations between social media use, mental health, and sleep outcomes.

	Subgroup	k	ES	95 % CI	Q	I^2	Difference (Q-test)
Social media use (SMU) and depression							
Region	Asia	12	0.075	0.064, 0.144	227.63	95.2	42.40**
	Europe	14	0.117	0.066, 0.169	170.72	89.3	
	North America	13	0.055	0.029, 0.080	10.66	0.0	
	Oceania	7	0.184	0.144, 0.224	42.40	16.3	
Design	Cross-sectional	35	0.090	0.059, 0.120	404.99	91.6	2.18
	Longitudinal	10	0.144	0.068, 0.218	115.22	92.2	
Depression measure	CESD	12	0.071	0.017, 0.124	91.02	87.9	5.42
	DASS-21	5	0.043	−0.070, 0.155	9.30	57.0	
	PHQ-9	5	0.104	−0.029, 0.233	23.65	83.1	
	Other	24	0.126	0.088, 0.165	207.23	88.9	
General vs specific type	Facebook	5	0.024	−0.042, 0.091	3.41	0.0	11.38*
	Instagram	3	0.110	−0.094, 0.306	3.75	46.7	
	Other specific	4	0.047	−0.149, 0.240	58.59	94.9	
	General	34	0.117	0.086, 0.148	266.03	87.6	
SMU measure	Validated	10	0.052	−0.030, 0.134	140.17	93.6	2.75
	Non-validated	36	0.116	0.088, 0.143	249.03	85.9	
Social media use (SMU) and anxiety							
Region	Asia	4	0.051	−0.064, 0.164	5.82	48.5	7.23*
	Europe	8	0.120	0.044, 0.194	161.72	95.7	
	North America	5	0.237	0.068, 0.393	26.56	84.9	
Design	Cross-sectional	14	0.136	0.064, 0.208	117.39	88.9	0.00
	Longitudinal	4	0.138	−0.014, 0.283	80.28	96.3	
Anxiety measure	DASS-21	5	0.055	−0.058, 0.166	10.72	62.7	4.58*
	Other	14	0.161	0.098, 0.222	209.08	93.8	
SMU measure	Valid	3	0.073	−0.248, 0.380	8.98	77.7	0.81
	Non-valid	16	0.145	0.086, 0.203	227.86	93.4	
Problematic social media use (PSMU) and depression							
Region	Asia	16	0.285	0.238, 0.330	93.39	83.9	0.47
	Europe	13	0.279	0.182, 0.371	218.32	94.5	
	North America	3	0.313	0.134, 0.473	10.18	80.4	
Design	Cross-sectional	30	0.297	0.252, 0.340	330.20	91.2	7.53**
	Longitudinal	3	0.200	0.077, 0.318	7.39	72.9	
Depression measure	CESD	9	0.268	0.199, 0.333	47.73	83.2	5.56
	DASS-21	5	0.353	0.273, 0.428	7.42	46.1	
	PHQ-9	6	0.313	0.247, 0.376	30.07	83.4	
	SDHS	4	0.286	0.111, 0.403	4.72	36.5	
	Other	9	0.263	0.111, 0.403	208.35	96.2	
General vs Specific PSMU	General	20	0.312	0.262, 0.360	267.64	92.9	0.74
	Facebook	11	0.282	0.227, 0.336	31.50	68.3	
PSMU measure	BFAS	6	0.275	0.178, 0.366	19.71	74.6	2.35
	BSMAS	6	0.331	0.263, 0.396	19.68	74.6	
	FIQ	3	0.307	0.016, 0.550	9.64	79.2	
	Other	18	0.278	0.208, 0.346	259.09	93.4	
Problematic social media use (PSMU) and anxiety							
Region	Asia	3	0.269	0.116, 0.410	39.87	90.0	46.67**
	Europe	5	0.359	0.283, 0.432	0.087	0.0	
	North America	3	0.217	0.171, 0.262	1.47	0.0	
Anxiety measure	DASS-21	4	0.350	0.274, 0.421	3.21	6.5	8.46**
	Other	8	0.236	0.160, 0.310	36.37	80.8	
General vs Specific PSMU	General	4	0.293	0.158, 0.417	19.20	84.4	0.00
	Facebook	6	0.296	0.207, 0.380	19.17	73.9	
PSMU measure	BFAS	6	0.278	0.186, 0.365	20.58	75.7	0.04
	Other	6	0.267	0.152, 0.345	39.42	87.3	
Problematic social media use (PSMU) and sleep quality							
Region	Asia	5	0.267	−0.020, 0.514	53.01	92.5	2.60
	Europe	3	0.032	−0.403, 0.454	39.25	94.9	
Sleep quality measure	PSQI	4	0.346	0.131, 0.530	21.95	86.3	7.41**
	Other	6	0.077	−0.116, 0.264	138.12	96.4	
General vs Specific PSMU	General	7	0.220	0.008, 0.413	186.03	96.8	0.64
	Facebook	3	0.108	−0.359, 0.532	54.21	96.3	

CESD = Centre for Epidemiological Studies-Depression; DASS-21 = Depression Anxiety and Stress Scale 21; PHQ-9 = *Patient Health Questionnaire-9*; SDHS = Short Depression-Happiness Scale; BFAS = Bergen Facebook Addiction Scale; BSMAS = Bergen Social Media Addiction Scale; FIQ = Facebook Intrusion Questionnaire; PSQI = Pittsburgh Sleep Quality Index.

* $p < .05$.

** $p < .01$.

Table 3
Summary of moderator analysis.

	Moderators				
	Mean age	Gender	Sample size	Publication year	Study quality
Social media use					
Depression	−0.010**	−0.001	<0.001	<0.001	−0.003
Anxiety	−0.007	−0.001	<0.001	−0.019	−0.022
Problematic social media use					
Depression	0.011	0.001	<0.001	0.003	0.006
Anxiety	0.0004	0.006*	<0.001	0.010	0.029
Wellbeing	−0.181***	−0.018***	0.001	−0.246***	−0.161***
Sleep quality	0.021	0.010	−0.0002	0.007	0.012

* $p < .05$.
** $p < .01$.
*** $p < .001$.

aggregate, there was a small and non-significant association between SMU and eating disorders ($\bar{r} = 0.125$, 95 % CI [−0.008, 0.258]) (Table 1 and Fig. S19). In sub-analyses excluding influential studies ($\bar{r} = 0.150$, 95 % CI [0.080, 0.219]), this association did not differ in direction and strength. (Table S6).

3.9. Social media use and sleep duration

A total of 17 studies assessed the association between SMU and sleep duration. Among studies, 12 were cross-sectional (70.6 % of 17 studies) and five were longitudinal (29.4 %). Among studies, 10 studies (58.8 %; representing n = 85,055 participants) reported significant cross-sectional associations between SMU and sleep duration [3,5,36,48,50,64,66,79,99,101], while one study ($n = 114$ participants) reported a non-significant association [124], and another study ($n = 290$ participants) reported that participants attributed delayed bedtime to SMU [58]. Two studies (11.7 %; n = 379,147 participants) reported significant longitudinal associations between SMU and later sleep duration [171,177], while three studies (17.6 %; $n = 1119$ participants) reported non-significant associations [151,156,169]. Two studies (11.7 %; n = 15,731 participants) reported that sleep duration mediated the association between SMU and depression [66,177].

Six studies (participant n = 14,417, median = 236, IQR = 2133.75, weighted mean age = 14.7 years, female = 50.5 %) could be included in the meta-analysis. In aggregate, there was a small and non-significant association between SMU and sleep duration ($\bar{r} = -0.062$, 95 % CI [−0.163, 0.039]) (Table 1 and Fig. S20). In sub-analyses excluding influential studies, this association ($\bar{r} = 0.022$, 95 % CI [−0.015, 0.058]) did not differ in strength (Table S6).

3.10. Social media use and sleep problems

A total of 23 studies assessed the association between SMU and sleep problems. Nineteen were cross-sectional (82.6 % of 23 studies) and four were longitudinal (17.4 %). Among studies, 16 studies (69.6 %; representing n = 58,261 participants) reported significant cross-sectional associations between SMU and sleep problems [2,5,6,36,37,48,50,66,70,72,74,85,98,115,128, 132], while three studies (13 %; n = 798 participants) reported non-significant associations [110,124,127]. Among longitudinal studies, only one study (4.3 %; n = 197 participants) reported a significant longitudinal association between SMU and later sleep problem [155], while three studies (13 %; n = 4404 participants) reported non-significant associations [169,172,173].

Eleven studies (participant n = 19,889, median = 702, IQR = 1106.5, weighted mean age = 15.5 years, female = 50.9 %) could be included in the meta-analysis. In aggregate, there was a small and non-significant association between SMU and sleep problems ($\bar{r} = 0.079$, 95 % CI [−0.037, 0.194]) (Table 1 and Fig. S21). In sub-analyses excluding influential studies, this association ($\bar{r} = 0.109$, 95 % CI [0.013, 0.205]) did not differ in strength or direction (Table S6).

3.11. Problematic social media use

Sixteen studies assessed the prevalence of problematic social media use (PSMU). Overall, the prevalence rate ranged between 2.5 % - 92.8 % [17,49]. Nine studies assessed the prevalence in Asian samples (ranging from 21.5 % to 57 %) [34,47,58,65,75,77,106,107,134], four studies in European samples (ranging from 2.5 % to 12.8 %) [17,19,78,121], two studies in North American samples (ranging from 17.9 % to 25.4 %) [46,107], one in an African sample (92.8 %) [49], and one in a Latin American sample (8.6 %) [126].

3.12. Problematic social media use and depression

A total of 46 studies assessed the association between PSMU and depression. Among studies, 38 were cross-sectional (82.6 % of 46 studies) and eight were longitudinal (17.4 %). Among studies, 35 (76.1 %; representing n = 27,358 participants) reported significant cross-sectional associations between PSMU and depression [16,17,22,27,29,32-35,43,45,46,54-57,65,73,75,80,88,89,97,100,105-107,113,118,119,121,122,127,129,134,135,168], while two studies (4.3 %; n = 1001 participants) reported non-significant associations [60,145]. Three studies (6.5 %; n = 9761 participants) reported co-morbidity between PSMU and depression (ranging between 18.1 % and 73.8 % among PSMU group) [49,106,107]. Five studies (10.9 %; n = 8697 participants) reported significant longitudinal associations between PSMU and later depression [143,160,162,165,176], while one study (2.2 %; $n = 397$ participants) detected a significant longitudinal association [166]. Moreover, two studies (4.3 %) found that depression predicted later PSMU [160,166], while two other studies (4.3 %) did not detect such effects [143,176].

Thirty-three studies (participant n = 23,966, median = 426, IQR = 319, weighted mean age = 18.6 years, female = 55.3 %) could be included in the meta-analysis. In aggregate, there was a moderate and significant association between PSMU and depression ($\bar{r} = 0.297$, 95 % CI [0.252, 0.341]) (Table 1 and Fig. S22). In sub-analyses excluding influential studies ($\bar{r} = 0.312$, 95 % CI [0.281, 0.345]), this association did not differ in strength and direction (Table S6).

Subgroup analysis results (Table 2, Figs. S23–S28) showed that the associations between SMU and depression differed significantly by design (cross-sectional vs longitudinal). On average studies utilizing cross-sectional design reported stronger association than studies utilized longitudinal design. However, the strength of association was not significantly different across region (Asia VS Europe vs North America), depression measures (CESD vs DASS-21 vs PHQ-9 vs SDHS vs others), social media platforms (general vs Facebook), and SMU measure (Bergen Facebook Addiction Scale [BFAS] (Andreassen et al., 2012) vs Bergen Social Media Addiction Scale [BSMAS] (Andreassen et al., 2016) vs Facebook Intrusion Questionnaire [FIQ] (Elphinston and Noller, 2011) vs others). Subgroup analyses also showed non-significant differences in the strength of the association between studies including samples aged between 12 and 30 and those including all ages (Table S7). No moderating effect of age, gender, sample size, publication year or study quality were detected (Table 3).

3.13. Problematic social media use and anxiety

A total of 16 studies assessed the association between PSMU and anxiety. Among studies, 14 were cross-sectional (87.5 % of 16 studies) and two were longitudinal (12.5 %). Almost all the studies (81.3 %; representing n = 11,257 participants) reported significant cross-sectional associations between PSMU and anxiety

[63,74,92,97,111,116,130,145,167,170,174,184,207]. Two studies (12.5 %; n = 9177 participants) reported comorbidity between PSMU and anxiety (ranging between 88.9 % and 93.9 % among PSMU group) [146–147].

Twelve studies (participant n = 10,020, median = 510, IQR = 166, weighted mean age = 20.6 years, female = 55.3 %) could be included in the meta-analysis. In aggregate, there was a small and significant association between PSMU and anxiety ($\bar{r}$ = 0.278, 95 % CI [0.221, 0.336]) (Table 1 and Fig. S29). In sub-analyses excluding influential studies ($\bar{r}$ = 0.280, 95 % CI [0.233, 0.328]), this association did not differ in strength and direction (Table S6).

Subgroup analysis results (Table 2, Figs. S30–S34) showed that the associations between PSMU and anxiety differed significantly by geographical region. On average studies from Europe reported stronger associations than studies based in Asia and North America. This association also significantly differed by anxiety measures, with studies assessing anxiety by DASS-21 reporting a stronger association between PSMU and anxiety than studies assessing anxiety by other scales. However, the strength of association was not significantly different across social media platforms (general vs Facebook), and SMU measure (BFAS vs others). Subgroup analyses also showed a significant difference in the strength of the association between studies including samples aged between 12 and 30 and those including all ages (Table S7). On average, studies including all ages sample reported stronger associations than studies including samples aged between 12 and 30 years. There was also a moderating effect of gender (Table 3) such that weaker associations between PSMU and anxiety were detected in female participant (coef. = 0.006, $p < .05$). However, no moderating effect of age, sample size, publication year or study quality were detected.

3.14. Problematic social media use and wellbeing

A total of three cross-sectional studies (participant n = 3342) assessed the association between PSMU and wellbeing and reported significant cross-sectional associations [126,135,180]. Two studies including three separate samples (participant n = 2842, weighted mean age = 19.6 years, female = 62.0 %) could be included in the meta-analysis. In aggregate, there was a small and significant association between PSMU and wellbeing ($\bar{r}$ = −0.246, 95 % CI [−0.352, −0.140]) (Table 1 and Fig. S35). The moderating effect of gender, age, publication year and study quality (Table 3) on the association between PSMU and wellbeing were detected.

3.15. Problematic social media use and sleep duration

A total of three studies assessed the association between PSMU and sleep duration. Among studies, two (66.6 % of 3 studies; representing n = 704 participants) reported significant cross-sectional associations between PSMU and sleep duration [75,97], while only one study (n = 981 participants) reported a non-significant association [38]. Three studies (participant n = 1685, weighted mean age = 20.9 years, female = 61.3 %) could be included in the meta-analysis. In aggregate, there was a small and non-significant association between PSMU and sleep duration ($\bar{r}$ = 0.038, 95 % CI [−0.010, 0.086]) (Table 1 and Fig. S36). No moderating effect of gender, age, sample size, publication year and study quality (Table 3) on the association between PSMU and sleep duration were detected.

3.16. Problematic social media use and sleep problems

A total of 15 studies assessed the association between PSMU and sleep problems. Thirteen studies were cross-sectional (86.7 % of 13) and two were longitudinal (13.3 %). All the cross-sectional studies (representing n = 14,416 participants) reported significant cross-sectional associations between PSMU and sleep problems [1,7,19,65,77,85,89,94,105,116,126,127,134]. Among longitudinal studies, one study (6.7 %; n = 874 participants) reported a significant longitudinal association between PSMU and later sleep problem and not vice-versa [176], while another study (6.7 %; n = 2021 participants) reported a non-significant association between PSMU and later sleep problems [172]. Two studies (one cross-sectional and one longitudinal) reported that sleep problem mediates the association between PSMU and depression [66,176].

Ten studies (participant n = 7967, median = 419, IQR = 748, weighted mean age = 16.5 years, female = 54.0 %) could be included in the meta-analysis. In aggregate, there was a small and significant association between PSMU and sleep problems ($\bar{r}$ = 0.189, 95 % CI [0.054, 0.323]) (Table 1 and Fig. S37). In sub-analyses excluding influential studies ($\bar{r}$ = 0.183, 95 % CI [0.066, 0.301]), this association did not differ in strength and direction (Table S6).

Subgroup analysis results (Table 2, Figs. S38–S40) showed that the associations between PSMU and sleep problems differed significantly by sleep problem measures (Pittsburgh Sleep Quality Index [PSQI] vs others scale) with studies assessing sleep problems using the PSQI reporting a stronger association between PSMU and sleep problems than studies assessing sleep problems with other scales. However, the strength of association was not significantly different across region (Asia vs Europe), and social media platforms (general vs Facebook). No moderating effect of age, gender, sample size, publication year or study quality were detected (Table 3).

3.17. Heterogeneity and publication bias

There was considerable heterogeneity among studies (Table 1) with I^2 ranging between 84.50 (PSMU and anxiety) and 97.06 (PSMU and sleep problems), except for the association between PSMU and sleep duration (0.00). The influence analysis identified several influential studies in the associations between SMU and depression (n = 5 studies), anxiety (n = 2 studies), wellbeing (n = 1 study), eating disorders (n = 1 study), sleep duration (n = 2 studies), and sleep problems (n = 3 studies) (Figs. S41–S46). Similarly, this analysis also identified several influential studies in the associations between PSMU and depression (n = 3 studies), anxiety (n = 2 studies), and sleep problems (n = 2 studies) (Figs. 47–49). Changes in the strength of associations after excluding influential studies was ranged between $\Delta\bar{r}$ = 0.002 [SMU and depression, and PSMU and anxiety] and $\Delta\bar{r}$ = 0.084 [SMU and sleep duration]).

Egger's test and funnel plots are presented in Table 1 and Figs. S50–S51 and do not indicate any publication bias. The Trim and fill analyses showed that the number of missing studies ranged between 0 and 8, indicating that over all analyses 16.67 % (ranging from 0 to 40.40 %) studies were missing, which suggests low scope of bias due to missing studies.

4. Discussion

The present review systematically synthesised the prior research investigating the associations between SMU/PSMU, mental health, and sleep. These associations can be categorized into those focused on SMU, and those focused on PSMU. The main results were: a) SMU was weakly but significantly associated with depression and anxiety, but not wellbeing or sleep problems; b) PSMU was weakly to moderately associated with depression, anxiety, wellbeing, and sleep problems; and c) Geographical regions, age, and gender were the most influential moderators of these associations. The weak associations detected between SMU and mental health may be attributable to the lack of sensitivity of the measures used, or the relatively "noisy" context of epidemiological studies which cannot account for all other environmental factors varying between participants. Alternatively, it may suggest that SMU alone may not be a major risk factor for poor mental health but may exacerbate the effects of pre-existing mental health problems, lower self-esteem, poor coping skills, etc. In contrast, the weak to moderate associations between PSMU, sleep problems and poor mental health, suggest that the

addictive component of SMU may be more likely to adversely impact these domains. Additionally, the associations between SMU/PSMU and mental health are not uniform across populations. Cultural norms and values, developmental stages, and gender-specific social dynamics are likely to influence these associations.

Our findings that SMU was weakly and PSMU was moderately associated with depression is consistent with previous meta-analyses (Cunningham et al., 2021; Huang, 2022; Ivie et al., 2020; Marino et al., 2018; Shannon et al., 2022; Vahedi and Zannella, 2021; Yang et al., 2022; Yoon et al., 2019). However, the current results extend beyond prior findings in showing a longitudinal association between baseline SMU and later depressive symptoms, but not vice versa. This is important as it sheds light, to our knowledge for the first time in a meta-analysis, on the directionality of the association between SMU and depression. Importantly, this finding does not support the notion of a bidirectional association between SMU and depression reported in past research (Lopes et al., 2022). This finding indicates that SMU may be a potential risk factor for depression. Findings from longitudinal studies also suggest that PSMU could be a risk factor for depression as these studies found an association between initial PSMU and later depressive symptoms (Li et al., 2018; Mu et al., 2020; Raudsepp, 2019). However, more longitudinal and experimental research is needed to confirm the directionality between SMU/PSMU and depression even though this study provides some support for such an associations.

In addition to depression, SMU was weakly and PSMU was moderately associated with anxiety. These findings are consistent with previous meta-analyses (Huang, 2022; Marino et al., 2018; Shannon et al., 2022; Wong et al., 2022). Furthermore, our findings on the associations between SMU/PSMU and anxiety also went beyond prior findings to show that in longitudinal studies baseline SMU had a significant positive association with later anxiety symptoms. This is not completely surprising as SMU/PSMU may lead to heightened fears of negative evaluations, cyberbullying victimization, and negative interactions which may result in increased anxiety. However, due to the lack of computable evidence, the opposite association (baseline anxiety predicting later SMU) and the bi-directional associations between PSMU and anxiety could not be assessed. Future longitudinal and experimental research is therefore needed to shed light on the directionality of these associations.

In addition to mental health, another unique contribution of the present study is that PSMU, but not SMU, was associated with sleep problems, although not with sleep duration. However, the directionality of these associations remains unclear due to the lack of sufficient longitudinal data. These findings are consistent with those of a previous systematic review (Alonzo et al., 2021), which showed that excessive SMU was associated with poor sleep quality while frequent SMU is a risk factor for sleep problems (Alonzo et al., 2021). Similarly, Li et al. (2017) found that PSMU was associated with a 119 % increase in insomnia symptoms. The origin of these effects is not fully understood. Increased social media engagement may be contributing to the development of sleep problems, or it may be a coping strategy whereby social media is used to deal with sleep problems. Alternatively, it is possible that depression, often associated with PSMU, promotes sleep problems, or alternatively that other complex interactions are the cause of disrupted sleep. Future research, preferably using a longitudinal design, is needed to tease out these alternative explanations by examining the interrelationship and directionality between SMU/PSMU, sleep problems, and mental health.

Another interesting finding is that associations between SMU/PSMU and mental health appear to differ across different geographical regions. Indeed, SMU was found to be associated with greater depressive symptomatology in Oceania compared to North America. On the other hand, SMU was found to be associated with greater anxiety symptoms in North America compared to Asia and Europe, while PSMU was associated with greater anxiety symptoms in Europe compared to Asia and North America. These results suggest that cultural norms and values could shape the way social media is used and perceived, and thus differently contributes to mental health. Indeed, previous studies reported significant differences in SMU (e.g., number and characteristics of friends, frequency and time spent on SMU) and personal factors related to SMU (e.g., self-disclosure, motivation for SMU, personality traits, loneliness, and shyness) by countries (Cho and Park, 2013; Jackson and Wang, 2013; Kalia et al., 2019). For example, social media users from Asian countries appear to spend less time, have lower number of online friends, and disclose less personal information on social media compared to social media users from Western countries. Therefore, future studies should seek to account for geographical differences while generalizing the findings and consider including geographical regions as a covariate to reduce bias linked to geographical and cultural differences. Additionally, future studies should also explore why the association between SMU/PSMU and anxiety symptoms are greater in some countries compared to others. Unfortunately, there was an insufficient number of studies from Africa and Latin America to compare these regions to the rest of the world. Similarly, Ghai et al. (2022) reviewed the studies that investigated the association between SMU and wellbeing in the Global South and also found a scarcity of studies from Sub-Saharan Africa and Latin America. Consequently, a greater research focus in these regions is needed. It is also worth noting that associations between PSMU, sleep, and mental health were not substantially different between social media platforms. However, the limited number of studies investigating different platforms with similar methodologies might have limited the ability to detect differences. Therefore, further research which contrasts different social media platforms is needed to clarify these questions.

Our findings that the association between SMU and depression and PSMU and anxiety was stronger in younger social media users is in contrast to previous meta-analyses that reported age as a non-significant moderator (Yang et al., 2022; Yin et al., 2018). The origin of these effects is not completely clear but may relate to the fact that young users tend to spend more time on social media (Kemp, 2023) and also tend to be more emotionally vulnerable due to SMU (Hayes et al., 2015). Similarly, the strength of the association between PSMU and anxiety was found to be stronger in female social media users. This could be attributable to several factors, including the higher amount of time spent on social media (Kemp, 2023), the documented higher prevalence rate of general anxiety in women (American Psychiatric Association, 2013), and a greater incidence of cyberbullying victimization against women (Kim et al., 2018). Further longitudinal research is needed to clarify which age groups are more vulnerable to PSMU and to confirm gender effects in the association between SMU/PSMU and mental health. This would allow for better-targeted interventions and therapeutic measures for at-risk users.

An interesting result of the present study (although non-significant) is that the strength of the association between SMU and depression appeared weaker than the association between anxiety and SMU. In contrast, PSMU had a greater association with depression compared to anxiety. Caution should be exercised in discussing non-significant differences, however it is interesting to consider them in the context of findings demonstrating differences in the neurobiological mechanisms and substrates underlying depression and anxiety. For example, a recent meta-analysis demonstrated that while depression symptomatology was negatively associated with the volume of several brain structures (e.g., hippocampus, putamen, pallidum and thalamus), anxiety tended to be positively associated with the volume of somewhat but not fully overlapping structures (e.g., amygdala) (Espinoza Oyarce et al., 2020).

4.1. Limitations of the study

The present study has several limitations. Firstly, 46.9 % of individual studies had incomplete information (e.g., age mean, gender proportion, sample size, and standard error of the regression coefficient), and thus could only be included in the systematic review nor in the meta-analysis. Future researchers should report this information

more consistently to better enable future reviews to synthesise the evidence base. Secondly, several moderation effects could not be tested for some associations due to insufficient data. This precluded systematic investigation of possible differences between different SMU platforms, though there is evidence such differences exist (e.g., the association between SMU and depression seems to vary by platform). This highlights the need for additional research to assess the associations between specific social media platform use, mental health, and sleep.

4.2. Future directions and practical implications

A key gap in our knowledge due to the lack of sufficient longitudinal data is related to the directionality of the associations between SMU/PSMU, sleep, and mental health, except for the association between SMU and depression. Thus, future longitudinal research including repeated measurement of these variables is needed to clarify these associations. In addition, it could also explore the contribution of underlying factors such as SMU motives and fear of missing out. Importantly, additional investigations should explore in depth the complex role of sleep in the associations between SMU and mental health as it is important for mental health professionals, educators, and other relevant professionals to consider the complex and varied impacts of PSMU on sleep and mental health. There is also a pressing need to identify the specific group of social media users who are more vulnerable to PSMU so interventions and therapeutic measures can be better targeted at those with greater need. In addition, educational programs should be developed to raise awareness of SMU's contribution to wellbeing and mental health and ensure healthy SMU.

5. Conclusion

This study confirms that SMU is associated with an increased risk of poor mental health and sleep problems, and also demonstrates stronger effects for PSMU than SMU. A key outstanding question relates to the directionality of the association between SMU/PSMU, mental health, and sleep. Future research should therefore use longitudinal designs and repeated measures to address this knowledge gap as such information is essential to develop interventions and preventive measures to mitigate the risk of SMU for the most vulnerable users.

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CRediT authorship contribution statement

Oli Ahmed: Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Erin I. Walsh:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Amy Dawel:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Khawlah Alateeq:** Writing – review & editing, Data curation. **Daniela Andrea Espinoza Oyarce:** Writing – review & editing, Data curation. **Nicolas Cherbuin:** Writing – review & editing, Writing – original draft, Supervision, Data curation, Conceptualization.

Declaration of competing interest

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Appendix A. Supplementary data

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References

Ahmed, O., Sultana, T., Alam, N., Griffiths, M.D., Hiramoni, F.A., 2022. Problematic social media use, personality traits, and mental health among Bangladeshi university students. Journal of Technology in Behavioral Science 7, 183–191. https://doi.org/10.1007/s41347-021-00235-1.

Alonzo, R., Hussain, J., Stranges, S., Anderson, K.K., 2021. Interplay between social media use, sleep quality, and mental health in youth: a systematic review. Sleep Med. Rev. 56, 101414 https://doi.org/10.1016/j.smrv.2020.101414.

American Psychiatric Association, 2013. Diagnostic and Statistical Manual of Mental Disorders: DSM-5™, 5th edition ed. American Psychiatric Publishing, Inc., Arlington, VA, US. doi:https://doi.org/10.1176/appi.books.9780890425596.

Andreassen, S.C., Pallesen, S., 2014. Social network site addiction - an overview. Curr. Pharm. Des. 20, 4053–4061. https://doi.org/10.2174/13816128113199990616.

Andreassen, C.S., Torsheim, T., Brunborg, G.S., Pallesen, S., 2012. Development of a Facebook addiction scale. Psychol. Rep. 110, 501–517. https://doi.org/10.2466/02.09.18.PR0.110.2.501-517.

Andreassen, C.S., Billieux, J., Griffiths, M.D., Kuss, D.J., Demetrovics, Z., Mazzoni, E., Pallesen, S., 2016. The relationship between addictive use of social media and video games and symptoms of psychiatric disorders: a large-scale cross-sectional study. Psychol. Addict. Behav. 30, 252–262. https://doi.org/10.1037/adb0000160.

Appel, M., Marker, C., Gnambs, T., 2020. Are social media ruining our lives? A review of Meta-analytic evidence. Rev. Gen. Psychol. 24, 60–74. https://doi.org/10.1177/1089268019880891.

Arias-de la Torre, J., Puigdomenech, E., García, X., Valderas, J.M., Eiroa-Orosa, F.J., Fernández-Villa, T., Molina, A.J., Martín, V., Serrano-Blanco, A., Alonso, J., Espallargues, M., 2020. Relationship between depression and the use of Mobile technologies and social media among adolescents: umbrella review. J. Med. Internet Res. 22, e16388 https://doi.org/10.2196/16388.

Balduzzi, S., Rücker, G., Schwarzer, G., 2019. How to perform a meta-analysis with R: a practical tutorial. Evidence-Based Mental Health and Quality of Life Outcomes 22, 153–160. https://doi.org/10.1136/ebmental-2019-300117.

Bányai, F., Zsila, Á., Király, O., Maraz, A., Elekes, Z., Griffiths, M.D., Andreassen, C.S., Demetrovics, Z., 2017. Problematic social media use: results from a large-scale nationally representative adolescent sample. PloS One 12, e0169839. https://doi.org/10.1371/journal.pone.0169839.

Best, P., Manktelow, R., Taylor, B., 2014. Online communication, social media and adolescent wellbeing: a systematic narrative review. Child Youth Serv. Rev. 41, 27–36. https://doi.org/10.1016/j.childyouth.2014.03.001.

Buysse, D.J., Reynolds, C.F., Monk, T.H., Berman, S.R., Kupfer, D.J., 1989. The Pittsburgh sleep quality index: a new instrument for psychiatric practice and research. Psychiatry Res. 28, 193–213. https://doi.org/10.1016/0165-1781(89)90047-4.

Carbonell, X., Panova, T., 2017. A critical consideration of social networking sites' addiction potential. Addict. Res. Theory 25, 48–57. https://doi.org/10.1080/16066359.2016.1197915.

Casale, S., Rugai, L., Fioravanti, G., 2018. Exploring the role of positive metacognitions in explaining the association between the fear of missing out and social media addiction. Addict. Behav. 85, 83–87. https://doi.org/10.1016/j.addbeh.2018.05.020.

Cherbuin, N., Kim, S., Anstey, K.J., 2015. Dementia risk estimates associated with measures of depression: a systematic review and meta-analysis. BMJ Open 5, e008853. https://doi.org/10.1136/bmjopen-2015-008853.

Cho, S.E., Park, H.W., 2013. A qualitative analysis of cross-cultural new media research: SNS use in Asia and the west. Qual. Quant. 47, 2319–2330. https://doi.org/10.1007/s11135-011-9658-z.

Crone, E.A., Konijn, E.A., 2018. Media use and brain development during adolescence. Nat. Commun. 9, 588. https://doi.org/10.1038/s41467-018-03126-x.

Cunningham, S., Hudson, C.C., Harkness, K., 2021. Social media and depression symptoms: a Meta-analysis. Research on Child and Adolescent Psychopathology 49, 241–253. https://doi.org/10.1007/s10802-020-00715-7.

Das-Friebel, A., Lenneis, A., Realo, A., Sanborn, A., Tang, N.K.Y., Wolke, W., von Mühlenen, A., Lemola, S., 2020. Bedtime social media use, sleep, and affective wellbeing in young adults: an experience sampling study. J. Child Psychol. Psychiatry Allied Discip. 61, 1138–1149. https://doi.org/10.1111/jcpp.13326.

Deters, F.G., Mehl, M.R., 2012. Does posting Facebook status updates increase or decrease loneliness? An online social networking experiment. Soc. Psychol. Personal. Sci. 4, 579–586. https://doi.org/10.1177/1948550612469233.

Diener, E., Emmons, R.A., Larsen, R.J., Griffin, S., 1985. The satisfaction with life scale. J. Pers. Assess. 49, 71–75. https://doi.org/10.1207/s15327752jpa4901_13.

Diener, E., Wirtz, D., Biswas-Diener, R., Tov, W., Kim-Prieto, C., Choi, D.-w., Oishi, S., 2009. New measures of well-being, In: Diener, E. (Ed.), Assessing Well-Being: The Collected Works of Ed Diener. Springer Netherlands, Dordrecht, pp. 247–266. doi: https://doi.org/10.1007/978-90-481-2354-4_12.

Duval, S., Tweedie, R., 2000. Trim and fill: a simple funnel-plot–based method of testing and adjusting for publication Bias in Meta-analysis. Biometrics 56, 455–463. https://doi.org/10.1111/j.0006-341X.2000.00455.x.
Egger, M., Smith, G.D., Schneider, M., Minder, C., 1997. Bias in meta-analysis detected by a simple, graphical test. BMJ 315, 629. https://doi.org/10.1136/bmj.315.7109.629.
Ellison, N.B., Steinfield, C., Lampe, C., 2007. The benefits of Facebook "friends": social capital and college students' use of online social network sites. J. Comput.-Mediat. Commun. 12, 1143–1168. https://doi.org/10.1111/j.1083-6101.2007.00367.x.
Elphinston, R.A., Noller, P., 2011. Time to face it! Facebook intrusion and the implications for romantic jealousy and relationship satisfaction. Cyberpsychol. Behav. Soc. Netw. 14, 631–635. https://doi.org/10.1089/cyber.2010.0318.
Espinoza Oyarce, D.A., Shaw, M.E., Alateeq, K., Cherbuin, N., 2020. Volumetric brain differences in clinical depression in association with anxiety: a systematic review with meta-analysis. J. Psychiatry Neurosci. 45, 406. https://doi.org/10.1503/jpn.190156.
Fairburn, C.G., Beglin, S.J., 1994. Assessment of eating disorders: interview or self-report questionnaire? Int. J. Eat. Disord. 16, 363–370. https://doi.org/10.1002/1098-108X(199412)16:4%3C363::AID-EAT2260160405%3E3.0.CO;2-%23.
Finserås, T.R., Hjetland, G.J., Sivertsen, B., Colman, I., Hella, R.T., Andersen, A.I.O., Skogen, J.C., 2023. Reexploring problematic social media use and its relationship with adolescent mental health. Findings from the "LifeOnSoMe"-study. Psychol. Res. Behav. Manag. 16, 5101–5111. https://doi.org/10.2147/PRBM.S435578.
Fumagalli, E., Shrum, L.J., Lowrey, T.M., 2024. The effects of social media consumption on adolescent psychological well-being. J. Assoc. Consum. Res. 9, 119–130. https://doi.org/10.1086/728739.
Ghai, S., Magis-Weinberg, L., Stoilova, M., Livingstone, S., Orben, A., 2022. Social media and adolescent well-being in the global south. Curr. Opin. Psychol. 46, 101318 https://doi.org/10.1016/j.copsyc.2022.101318.
Gradisar, M., Gardner, G., Dohnt, H., 2011. Recent worldwide sleep patterns and problems during adolescence: a review and meta-analysis of age, region, and sleep. Sleep Med. 12, 110–118. https://doi.org/10.1016/j.sleep.2010.11.008.
Griffiths, M., 2005. A 'components' model of addiction within a biopsychosocial framework. J. Subst. Abus. 10, 191–197. https://doi.org/10.1080/14659890500114359.
Harrer, M., Cuijpers, P., Furukawa, T.A., Ebert, D.D., 2022. Doing Meta-Analysis with R: A Hands-on Guide, 1st ed. Chapman & Hall/CRC Press, Boca Raton, FL and London.
Hayes, M., van Stolk-Cooke, K., Muench, F., 2015. Understanding Facebook use and the psychological affects of use across generations. Computers in Human Behavior 49, 507–511. https://doi.org/10.1016/j.chb.2015.03.040.
Higgins, J.P.T., Thomas, J., Chandler, J., Cumpston, M., Li, T., Page, M.J., Welch, V.A., 2019. Cochrane Handbook for Systematic Reviews of Interventions. John Wiley & Sons, Chichester, UK.
Huang, C., 2022. A meta-analysis of the problematic social media use and mental health. Int. J. Soc. Psychiatry 68, 12–33. https://doi.org/10.1177/0020764020978434.
Ivie, E.J., Pettitt, A., Moses, L.J., Allen, N.B., 2020. A meta-analysis of the association between adolescent social media use and depressive symptoms. J. Affect. Disord. 275, 165–174. https://doi.org/10.1016/j.jad.2020.06.014.
Jackson, L.A., Wang, J.-L., 2013. Cultural differences in social networking site use: a comparative study of China and the United States. Comput. Hum. Behav. 29, 910–921. https://doi.org/10.1016/j.chb.2012.11.024.
Kalia, P., Ben Dahmane Mouelhi, N., Tebessi Hachana, S., Malek, F., Dahmen, M., 2019. Using social networking sites: A qualitative cross-cultural comparison. In: Somani, A. K., Ramakrishna, S., Chaudhary, A., Choudhary, C., Agarwal, B. (Eds.), Emerging Technologies in Computer Engineering: Microservices in Big Data Analytics. Springer Singapore, Singapore, pp. 269–285. https://doi.org/10.1007/978-981-13-8300-7_23.
Kaur, P., Dhir, A., Alkhalifa, A.K., Tandon, A., 2021. Social media platforms and sleep problems: a systematic literature review, synthesis and framework for future research. Internet Res. 31, 1121–1152. https://doi.org/10.1108/INTR-04-2020-0187.
Keles, B., McCrae, N., Grealish, A., 2020. A systematic review: the influence of social media on depression, anxiety and psychological distress in adolescents. Int. J. Adolesc. Youth 25, 79–93. https://doi.org/10.1080/02673843.2019.1590851.
Kelly, Y., Zilanawala, A., Booker, C., Sacker, A., 2018. Social media use and adolescent mental health: findings from the UK millennium cohort study. EClinicalMedicine 6, 59–68. https://doi.org/10.1016/j.eclinm.2018.12.005.
Kemp, S., 2023. Digital 2023: Global Overview Report.
Kim, S., Kimber, M., Boyle, M.H., Georgiades, K., 2018. Sex differences in the association between cyberbullying victimization and mental health, substance use, and suicidal ideation in adolescents. Can. J. Psychiatry 64, 126–135. https://doi.org/10.1177/0706743718777397.
Kroenke, K., Spitzer, R.L., Williams, J.B.W., 2001. The PHQ-9. J. Gen. Intern. Med. 16, 606–613. https://doi.org/10.1046/j.1525-1497.2001.016009606.x.
Li, J.-B., Lau, J.T.F., Mo, P.K.H., Su, X.-F., Tang, J., Qin, Z.-G., Gross, D.L., 2017. Insomnia partially mediated the association between problematic internet use and depression among secondary school students in China. J. Behav. Addict. 6, 554–563. https://doi.org/10.1556/2006.6.2017.085.
Li, J.B., Mo, P.K.H., Lau, J.T.F., Su, X.F., Zhang, X., Wu, A.M.S., Mai, J.C., Chen, Y.X., 2018. Online social networking addiction and depression: the results from a large-scale prospective cohort study in Chinese adolescents. J. Behav. Addict. 7, 686–696. https://doi.org/10.1556/2006.7.2018.69.
Liu, M., Kamper-DeMarco, K.E., Zhang, J., Xiao, J., Dong, D., Xue, P., 2022. Time spent on social media and risk of depression in adolescents: a dose-response Meta-analysis. Int. J. Environ. Res. Public Health 19, 5164. https://doi.org/10.3390/ijerph19095164.
Lopes, L.S., Valentini, J.P., Monteiro, T.H., Costacurta, M.C.d.F., Soares, L.O.N., Telfar-Barnard, L., Nunes, P.V., 2022. Problematic social media use and its relationship with depression or anxiety: a systematic review. Cyberpsychol. Behav. Soc. Netw. 25, 691–702. doi:https://doi.org/10.1089/cyber.2021.0300.
Lovibond, P.F., Lovibond, S.H., 1995. The structure of negative emotional states: comparison of the depression anxiety stress scales (DASS) with the Beck depression and anxiety inventories. Behav. Res. Ther. 33, 335–343. https://doi.org/10.1016/0005-7967(94)00075-U.
Ma, J., 2022. A Meta-Analysis of Social Media Usage with Stress, Anxiety, and Depression, Proceedings of the 2021 International Conference on Intelligent Medicine and Health (ICIMH '21). Association for Computing Machinery, New York, NY, USA, pp. 125–129. https://doi.org/10.1145/3484377.3487041.
Marino, C., Gini, G., Vieno, A., Spada, M.M., 2018. The associations between problematic Facebook use, psychological distress and well-being among adolescents and young adults: a systematic review and meta-analysis. J. Affect. Disord. 226, 274–281. https://doi.org/10.1016/j.jad.2017.10.007.
Marteau, T.M., Bekker, H., 1992. The development of a six-item short-form of the state scale of the Spielberger state—trait anxiety inventory (STAI). Br. J. Clin. Psychol. 31, 301–306. https://doi.org/10.1111/j.2044-8260.1992.tb00997.x.
McCrae, N., Gettings, S., Purssell, E., 2017. Social media and depressive symptoms in childhood and adolescence: a systematic review. Adolesc. Res. Rev. 2, 315–330. https://doi.org/10.1007/s40894-017-0053-4.
Morales-Muñoz, I., Gregory, A.M., 2023. Sleep and mental health problems in children and adolescents. Sleep Med. Clin. 18, 245–254. https://doi.org/10.1016/j.jsmc.2023.01.006.
Mu, W., Zhu, D., Wang, Y., Li, F., Ye, L., Wang, K., Zhou, M., 2020. Three-wave longitudinal survey on the relationship between neuroticism and depressive symptoms of first-year college students: addictive use of social media as a moderated mediator. Int. J. Environ. Res. Public Health 17, 1–13. https://doi.org/10.3390/ijerph17176074.
Nasirudeen, A.M.A., Lee Chin Adeline, L., Wat Neo Josephine, K., Lay Seng, L., Wenjie, L., 2017. Impact of social media usage on daytime sleepiness: a study in a sample of tertiary students in Singapore. DIGITAL HEALTH 3, 2055207617699766. https://doi.org/10.1177/2055207617699766.
National Sleep Foundation, 2011. Sleep in America Poll: Poll, Technology and Sleep. National Sleep Foundation, Washington DC, USA.
O'Keeffe, G.S., Clarke-Pearson, K., Council on, C., Media, 2011. The impact of social media on children, adolescents, and families. Pediatrics 127, 800–804. https://doi.org/10.1542/peds.2011-0054.
Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., Shamseer, L., Tetzlaff, J.M., Akl, E.A., Brennan, S.E., Chou, R., Glanville, J., Grimshaw, J.M., Hróbjartsson, A., Lalu, M.M., Li, T., Loder, E.W., Mayo-Wilson, E., McDonald, S., McGuinness, L.A., Stewart, L.A., Thomas, J., Tricco, A.C., Welch, V.A., Whiting, P., Moher, D., 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 372, n71. https://doi.org/10.1136/bmj.n71.
Pellegrino, A., Stasi, A., Bhatiasevi, V., 2022. Research trends in social media addiction and problematic social media use: a bibliometric analysis. Front. Psych. 13, 1017506. https://doi.org/10.3389/fpsyt.2022.1017506.
Piteo, E.M., Ward, K., 2020. Review: social networking sites and associations with depressive and anxiety symptoms in children and adolescents – a systematic review. Child Adolesc. Mental Health 25, 201–216. https://doi.org/10.1111/camh.12373.
Posit team, 2023. Studio: Integrated Development Environment for R. Posit Software, PBC, Boston, MA.
R Core Team, 2021. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.
Radloff, L.S., 1977. The CES-D scale: a self-report depression scale for research in the general population. Appl. Psychol. Measur. 1, 385–401. https://doi.org/10.1177/014662167700100306.
Raudsepp, L., 2019. Brief report: problematic social media use and sleep disturbances are longitudinally associated with depressive symptoms in adolescents. J. Adolesc. 76, 197–201. https://doi.org/10.1016/j.adolescence.2019.09.005.
Rodgers, R.F., Slater, A., Gordon, C.S., McLean, S.A., Jarman, H.K., Paxton, S.J., 2020. A biopsychosocial model of social media use and body image concerns, disordered eating, and muscle-building behaviors among adolescent girls and boys. J. Youth Adolesc. 49, 399–409. https://doi.org/10.1007/s10964-019-01190-0.
Sanders, T., Noetel, M., Parker, P., Del Pozo Cruz, B., Biddle, S., Ronto, R., Hulteen, R., Parker, R., Thomas, G., De Cocker, K., Salmon, J., Hesketh, K., Weeks, N., Arnott, H., Devine, E., Vasconcellos, R., Pagano, R., Sherson, J., Conigrave, J., Lonsdale, C., 2024. An umbrella review of the benefits and risks associated with youths' interactions with electronic screens. Nat. Hum. Behav. 8, 82–99. https://doi.org/10.1038/s41562-023-01712-8.
Santos, R.M.S., Mendes, C.G., Sen Bressani, G.Y., de Alcantara Ventura, S., de Almeida Nogueira, Y.J., de Miranda, D.M., Romano-Silva, M.A., 2023. The associations between screen time and mental health in adolescents: a systematic review. BMC Psychology 11, 127. https://doi.org/10.1186/s40359-023-01166-7.
Schønning, V., Hjetland, G.J., Aarø, L.E., Skogen, J.C., 2020. Social media use and mental health and well-being among adolescents – a scoping review. Front. Psychol. 11, 01949. https://doi.org/10.3389/fpsyg.2020.01949.
Seabrook, E.M., Kern, M.L., Rickard, N.S., 2016. Social networking sites, depression, and anxiety: a systematic review. JMIR Ment Health 3, e50. https://doi.org/10.2196/mental.5842.
Shannon, H., Bush, K., Villeneuve, P.J., Hellemans, K.G.C., Guimond, S., 2022. Problematic social media use in adolescents and young adults: systematic review and Meta-analysis. JMIR Ment Health 9, e33450. https://doi.org/10.2196/33450.
Shensa, A., Escobar-Viera, C.G., Sidani, J.E., Bowman, N.D., Marshal, M.P., Primack, B. A., 2017. Problematic social media use and depressive symptoms among U.S. young

adults: a nationally-representative study. Soc Sci Med 182, 150–157. https://doi.org/10.1016/j.socscimed.2017.03.061.

Ulvi, O., Karamehic-Muratovic, A., Baghbanzadeh, M., Bashir, A., Smith, J., Haque, U., 2022. Social media use and mental health: a global analysis. Epidemiologia 11–25. https://doi.org/10.3390/epidemiologia3010002.

Vahedi, Z., Zannella, L., 2021. The association between self-reported depressive symptoms and the use of social networking sites (SNS): a meta-analysis. Curr. Psychol. 40, 2174–2189. https://doi.org/10.1007/s12144-019-0150-6.

Valkenburg, P.M., Meier, A., Beyens, I., 2022. Social media use and its impact on adolescent mental health: an umbrella review of the evidence. Curr. Opin. Psychol. 44, 58–68. https://doi.org/10.1016/j.copsyc.2021.08.017.

van den Eijnden, R.J.J.M., Geurts, S.M., Ter Bogt, T.F.M., van der Rijst, V.G., Koning, I. M., 2021. Social media use and adolescents' sleep: a longitudinal study on the protective role of parental rules regarding internet use before sleep. Int. J. Environ. Res. Public Health 18, 1–13. https://doi.org/10.3390/ijerph18031346.

Viechtbauer, W., 2010. Conducting Meta-analyses in R with the metafor package. J. Stat. Softw. 36, 1–48. https://doi.org/10.18637/jss.v036.i03.

Watson, D., Clark, L.A., Tellegen, A., 1988. Development and validation of brief measures of positive and negative affect: the PANAS scales. J. Pers. Soc. Psychol. 54, 1063–1070. https://doi.org/10.1037/0022-3514.54.6.1063.

Wells, G., Shea, B., O'Connell, D., Robertson, J., Peterson, J., Welch, V., Losos, M., Tugwell, P., 2013. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. www.ohri.ca/programs/clinical_epidemiology/oxford.asp.

Wong, H.Y., Mo, H.Y., Potenza, M.N., Chan, M.N.M., Lau, W.M., Chui, T.K., Pakpour, A. H., Lin, C.Y., 2020. Relationships between severity of internet gaming disorder, severity of problematic social media use, sleep quality and psychological distress. Int. J. Environ. Res. Public Health 17. https://doi.org/10.3390/ijerph17061879.

Wong, J., Yi, P.X., Quek, F.Y.X., Lua, V.Y.Q., Majeed, N.M., Hartanto, A., 2022. A four-level meta-analytic review of the relationship between social media and well-being: a fresh perspective in the context of COVID-19. Curr. Psychol. https://doi.org/10.1007/s12144-022-04092-w.

Yang, Q., Liu, J., Rui, J., 2022. Association between social network sites use and mental illness: a meta-analysis. Cyberpsychology: Journal of Psychosocial Research on Cyberspace 16, Article 1. https://doi.org/10.5817/CP2022-1-1.

Yin, X.-Q., de Vries, D.A., Gentile, D.A., Wang, J.-L., 2018. Cultural background and measurement of usage moderate the association between social networking sites (SNSs) usage and mental health: a Meta-analysis. Soc. Sci. Comput. Rev. 37, 631–648. https://doi.org/10.1177/0894439318784908.

Yoon, S., Kleinman, M., Mertz, J., Brannick, M., 2019. Is social network site usage related to depression? A meta-analysis of Facebook–depression relations. J. Affect. Disord. 248, 65–72. https://doi.org/10.1016/j.jad.2019.01.026.